

Chapter 18: Fetal Disorders — Revision Notes

Williams Obstetrics 26e, pp. 904–937 of the PDF. High-yield numbers are in **bold**. The four tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 5 figures are included from the PDF. The Kleihauer-Betke formula (also missing from the text) is taken from the PDF page.

Big picture: Fetal anemia (alloimmune, infectious, genetic or from fetomaternal hemorrhage) is detected by **MCA peak systolic velocity** and treated by **intrauterine transfusion**. Anti-D immune globulin prevents most D alloimmunization. Platelet alloimmunization (FNAIT) can hit the **first** pregnancy. Hydrops is the final common pathway of many disorders; **≥90% is nonimmune**.

1. Fetal Anemia

- **Causes:**
 - **Red cell alloimmunization** → overproduction of immature red cells (erythroblastosis fetalis), now called **hemolytic disease of the fetus and newborn (HDFN)**
 - **Infection**, particularly **parvovirus B19**
 - **α⁰-thalassemia**: a common cause of severe anemia and nonimmune hydrops in **Southeast Asian** populations
 - Rare genetic causes: production disorders (Diamond-Blackfan, Fanconi), enzymopathies (G6PD, pyruvate kinase deficiency), membrane defects (spherocytosis, elliptocytosis), lysosomal storage diseases (Gaucher, Niemann-Pick, MPS VII), myeloproliferative disorders (leukemias)
 - **Fetomaternal hemorrhage**
- Detected by Doppler of the **MCA peak systolic velocity**.
- Progressive anemia → heart failure → **hydrops** → death.
- **Intrauterine transfusion survival >90%**; even with hydrops it approaches **80%**.

2. Red Cell Alloimmunization

- **36 blood group systems** and **360 erythrocyte antigens** are recognized.
- The fetus inherits a paternal antigen that the mother lacks; enough fetal cells in her circulation → sensitization → maternal IgG crosses the placenta → hemolysis.
- **Why alloimmunization is uncommon:**
 1. Low prevalence of incompatible antigens
 2. Insufficient transplacental passage of fetal antigens and maternal antibodies
 3. **Maternal-fetal ABO incompatibility** → rapid clearance of fetal cells before an immune response
 4. Variable antigenicity
 5. Variable maternal immune response
- Prevalence in pregnancy **~1%**.
- Most severe anemia needing antenatal transfusion: **anti-D, anti-Kell, anti-c, anti-E**.
- **Screening:** blood type + antibody screen at the first prenatal visit (**indirect Coombs** detects unbound antibody). If positive → identify antibody, IgG vs IgM, and titer.
 - **Only IgG matters** (IgM does not cross the placenta).
- **Critical titer** = level at which significant fetal anemia may develop: usually **1:8 to 1:32**. If the lab's anti-D cutoff is 1:16, a titer **≥1:16** means possible severe disease.
 - **Exception: Kell** (titer does not predict severity).

Table 18-1. Selected Red Cell Antigens and Their Relationship to Fetal Hemolytic Disease

Blood Group System	Antigens	Fetal Hemolysis Potential
CDE (Rh)	D, c	Severe disease risk
	E, Be ^a , Ce, Cw, Cx, ce, Dw, Evans, e, G, Goa7, Hr, Hro, JAL, HOFM, LOCR, Riv, Rh29, Rh32, Rh42, Rh46, STEM, Tar	Severe disease infrequent, mild disease risk
Kell	K	Severe disease risk
	k, Kp ^a , Kp ^b , K11, K22, Ku, Js ^a , Js ^b , Ula	Severe disease infrequent, mild disease risk
Duffy	Fy ^a	Severe disease infrequent, mild disease risk
	Fy^b	Not associated with fetal hemolytic disease
Kidd	JK ^a	Severe disease infrequent, mild disease risk
	JK ^b , JK ³	Mild disease possible
MNS	M, N, S, s, U, Mt ^a , Ena, Far, Hil, Hut, Mi ^a , Mit, Mut, Mur, Mv, sD, Vw	Severe disease infrequent, mild disease risk
Colton	Co ^a , Co ³	Severe disease infrequent, mild disease risk
Diego	Di ^a , Di ^b , Wr ^a , Wr ^b	Severe disease infrequent, mild disease risk
Dombrock	Do ^a , Gy ^a , Hy, Jo ^a	Mild disease possible
Gerbich	Ge ² , Ge ³ , Ge ⁴ , Ls ^a	Mild disease possible
Scianna	Sc2	Mild disease possible
I	I, i	Not associated with fetal hemolytic disease
Lewis	Le^a, Le^b	Not associated with fetal hemolytic disease

Data from de Haas, 2015; Moise, 2008; Weinstein, 1982.

Memory hook: "Lewis Lives, I Is fine, Duffy B Doesn't Bother": Lewis, I and Fy^b are the harmless ones. **Lewis and I** are IgM cold agglutinins not expressed on fetal red cells.

CDE (Rh) incompatibility

- Five antigens: **C, c, D, E, e**. There is **no "d" antigen**; D-negative = absence of D. **>200 D variants** exist. "Rhesus" is no longer used in transfusion medicine.
- Sensitization after as little as 0.1 mL** of fetal red cells.
- Genes **RHD and RHCE** on the **short arm of chromosome 1**, inherited together.
- D-positive**: ~85% of non-Hispanic white, ~90% Native American, 93% African American and Hispanic, **99% Asian** Americans.
- D alloimmunization complicates **0.5–0.9%** of pregnancies.
- Without prophylaxis**, a D-negative woman delivering a D-positive, **ABO-compatible** baby → **16%** alloimmunization:
 - 2%** by delivery, **7%** by 6 months postpartum, **7%** "sensitized" (antibodies only in the next pregnancy)
 - With **ABO incompatibility** the risk falls to **2%**.
- Also after 1st-trimester complications, prenatal procedures and trauma (Table 18-2).

Table 18-2. Causes of Fetomaternal Hemorrhage Associated with Red Cell Antigen Alloimmunization^a

Pregnancy Loss	Procedures	Other
Abortion, spontaneous or elective	Amniocentesis	Antepartum bleeding, including threatened abortion
Ectopic pregnancy	Chorionic villus sampling	Delivery, vaginal or cesarean
Fetal death (any trimester)	Fetal blood sampling	External cephalic version
	Molar pregnancy evacuation	Placental abruption
		Trauma to the abdomen during pregnancy

^aFor each of the above, anti-D immune globulin is recommended. Expanded from American Academy of Pediatrics, 2017; ACOG, 2019b.

C, c, E and e antigens

- Less immunogenic than D, but can cause hemolytic disease.
- Sensitization in **~0.3%** of pregnancies = **~30%** of red cell alloimmunization cases.
- **Anti-E is the most common**, but **anti-c more often needs fetal/neonatal transfusion** than anti-E or anti-C.

The grandmother effect

- Small amounts of maternal blood enter the fetus in virtually all pregnancies.
- A **D-negative female fetus** exposed to her **D-positive mother's** cells may become sensitized → makes anti-D before or during her own pregnancy. Her fetus is then jeopardized by antibodies provoked by the **grandmother's** red cells.

Minor antigens

- Because anti-D prophylaxis works, **proportionately more HDFN is now due to non-D ("minor") antigens**.
- **Kell** is among the most frequent. Others that can be severe: **Duffy A (Fy^a), MNS, Kidd (Jk^a)**.
- Most minor-antigen sensitization comes from an **incompatible blood transfusion**.
- If an IgG antibody is found and its significance is in doubt → **evaluate for hemolytic disease**.
- **No fetal risk: Lewis (Le^a, Le^b) and I** (IgM cold agglutinins, not on fetal cells); **Duffy B (Fy^b)**.

Kell alloimmunization

- **~90%** of non-Hispanic white and up to **98%** of African Americans are **Kell negative**. Kell type is not routinely checked.
- **~90% of Kell sensitization is from transfusion** of Kell-positive blood → take a transfusion history.
- **More rapid and more severe:** anti-Kell attacks **erythrocyte precursors in the marrow** → suppressed erythropoiesis rather than hemolysis → **titer does not predict severity**.
- Sloatweg (2018): titer **1:4** had **100% sensitivity**, 27% specificity, 60% PPV for transfusion need. **>50%** of Kell-positive fetuses needed transfusion.
- **ACOG: do not use antibody titers to monitor Kell-sensitized pregnancies** (surveil by MCA Doppler regardless of titer).

ABO incompatibility

- **Most common cause of hemolytic disease in newborns**, but **no appreciable fetal hemolysis**:
 - Anti-A/anti-B are mostly **IgM** (do not cross)
 - Fetal cells have **fewer A and B sites**
- **~20%** of newborns are ABO incompatible; only **5%** are affected clinically, and anemia is mild.

- **Differs from CDE disease:**
 - Often affects the **firstborn** (group O women have anti-A/anti-B isoagglutinins from bacterial exposure)
 - **Rarely worsens** in successive pregnancies
 - **No fetal surveillance or early delivery** needed
- Postnatal: watch for **hyperbilirubinemia** (phototherapy, occasionally transfusion).

3. Management of the Alloimmunized Pregnancy

- D-alloimmunized fetuses: **25–30%** mild–moderate anemia; up to **25%** severe enough to cause **hydrops**.
- **Titer below critical value → repeat every 4 weeks.**
- **Once the critical titer is reached, no benefit in repeating it** — the pregnancy stays at risk even if the titer falls → further evaluation.
- **Prior affected pregnancy** → at risk **regardless of titer.**

Fetal risk assessment

- Anti-D shows maternal sensitization, not fetal status. **Up to 40%** of D-negative women carry a **D-negative fetus**.
 - Titer can rise with a D-negative fetus (**amnestic response**).
- Non-Hispanic white D-negative woman: partner is D-positive **85%**; of these, **60% heterozygous** → only half the children at risk.
- **Step 1: paternal antigen status.** If paternity is certain and the father is antigen-negative → **not at risk**.
- **Father D-positive → paternal zygosity** by DNA analysis. If **heterozygous or paternity unknown → fetal genotype**:
 - **Amniocentesis + PCR of uncultured amniocytes: PPV 100%, NPV ~97%.** Also for E/e, C/c, Duffy, Kell, Kidd, M/N.
 - **CVS NOT recommended** (↑ fetomaternal hemorrhage → worsens alloimmunization).
 - **cfDNA fetal RHD genotyping** (maternal plasma): **sensitivity >99%, specificity >95%.** Routine in parts of Europe, not yet widely in the US.
 - Uses: (1) in sensitized women, find D-negative fetuses that need no surveillance; (2) in unsensitized women, withhold anti-D if the fetus is D-negative.
 - **ACOG does not recommend routine cfDNA** screening in D-negative women until shown to be cost-effective.
- **Standard pathway:** titer surveillance → once **critical titer** → serial **MCA PSV** → **fetal blood sampling ± transfusion** if MCA PSV exceeds the severe-anemia threshold.
- **Amnionic fluid ΔOD450 (bilirubin) is no longer recommended.**
- **Maternal IVIG** in severe early disease: delays the first transfusion (to beyond **20 wk**) by an average of **3 weeks** and ↓ hydrops. Mechanism unclear.

MCA Doppler velocimetry

- **Recommended test to detect fetal anemia** (SMFM).
- Anemic fetus shunts blood to the brain; velocity ↑ from ↑ **cardiac output** + ↓ **blood viscosity**.
- **Mari (2000):** threshold **1.5 MoM** detected **all** moderate–severe anemia: **sensitivity 100%, false-positive 12%.**
- **1.0–1.5 MoM and rising → at least weekly** Doppler.
- False-positives ↑ **beyond 34 wk** (normal ↑ cardiac output). At Parkland, MCA PSV is **not measured beyond 35 wk** (hydrops surveillance instead).

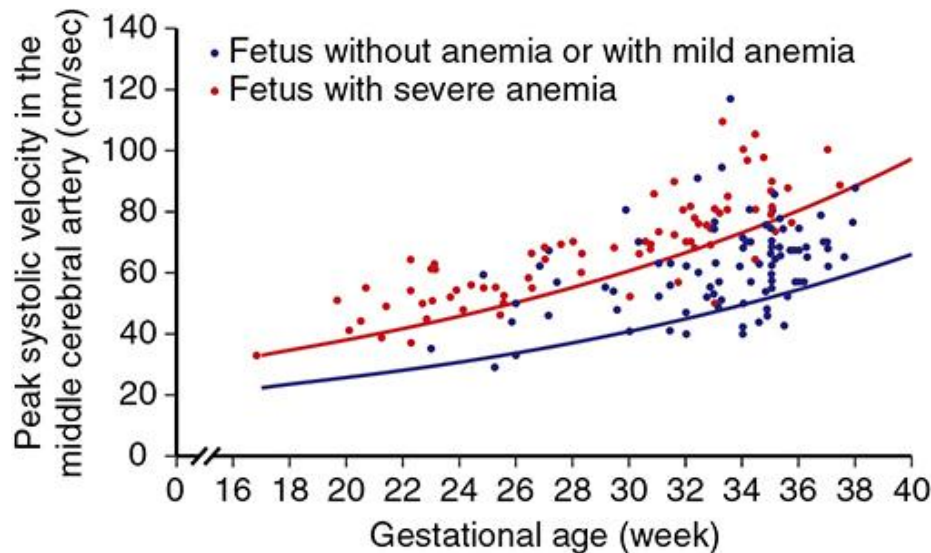


Figure 18-1. MCA peak systolic velocity vs gestational age: severely anemic fetuses (red dots) plot above the 1.5 MoM line (red).

Fetal blood transfusion

- **Indication:** MCA PSV **>1.5 MoM**, or hydrops with anemia as the leading cause → fetal blood sampling ± transfusion.
- Usually done **before 34–35 wk**; later, delivery risks are usually lower than transfusion risks.
- **Transfuse only if fetal Hct <30%**. With hydrops, Hct is usually **≤15%**.
- **Route:** usually **intravascular** (umbilical vein). **Intraperitoneal** if the vein is too narrow early in the 2nd trimester. Hydrops impairs peritoneal absorption → some do both.

Donor red cells are...	Why
Type O, D-negative	Compatible
CMV-negative	Prevent infection
Packed to Hct ~80%	Prevent volume overload
Irradiated	Prevent fetal graft-vs-host
Leukocyte-poor	—

- **Vecuronium** (paralytic) may be given to the fetus to stop movement.
- **Target Hct 40–50%** (nonhydropic fetus).
- **Volume (mL) $\approx$ EFW (g) $\times$ 0.02 for each 10% rise in Hct** needed.
- Severely anemic fetus at **18–24 wk**: smaller first volume, repeat in **~2 days**. Then every **2–4 weeks**.
- **After a transfusion, the MCA threshold rises to 1.70 MoM** (adult donor cells have smaller MCV).
- Post-transfusion **Hct falls ~1%/day** (faster with hydrops).

Outcomes

- **Overall survival ~95%**.
- Complications: **fetal death 2%**, emergent cesarean 1%, infection 0.3%, PPROM 0.3%.
- **Stillbirth >15% if transfusion needed before 20 wk**.
- Hydropic fetuses: neonatal survival **~80%**; **95%** if hydrops resolves vs **<40%** if it persists.
- **LOTUS (>450 pregnancies):** anti-D 80%, anti-Kell 12%, anti-c 5%; **~¼ hydropic**; **>½ needed neonatal exchange transfusion**; **<5% severe neurodevelopmental impairment**.

4. Prevention of Anti-D Alloimmunization

- Used for >50 years, yet **50% of women worldwide** who need it do not get it. Without it, up to **10%** of D-negative pregnancies develop HDFN.
- With prophylaxis, risk falls to **<0.2%**. Mechanism not fully understood.
- **Delivery hemorrhage causes up to 90%** of cases.
- **Postpartum dose within 72 h** → ↓ alloimmunization by **90%**.
- **28-wk dose** → 3rd-trimester alloimmunization **2% → 0.1%**.
- **"When in doubt, give it."** It does no harm if unneeded.

Preparations

Preparation	Route	Note
Cold ethanol fractionation + ultrafiltration	IM only	Contains plasma proteins → anaphylaxis if IV
Ion-exchange chromatography	IM or IV	IV useful for large fetomaternal hemorrhage

- Both remove viruses (hepatitis, HIV).
- **Half-life 16–24 days** → why it is given both at 28 wk and after delivery.

Dosing and timing (US)

- **Standard dose 300 µg (1500 IU) IM** covers **30 mL fetal whole blood = 15 mL fetal red cells**.
- **All D-negative unsensitized women at ~28 wk**; **repeat antibody screen before** the 28-wk dose.
- **Second dose within 72 h of delivery if the newborn is D-positive** (40% of newborns of D-negative women are D-negative).
- **Missed dose** → **give as soon as recognized** (some protection up to **28 days**).
- After sensitizing events (Table 18-2). **1st trimester: 50 or 120 µg** may suffice.
- **Weakly positive indirect Coombs (1:1 to 1:4)** after anti-D is harmless — not alloimmunization.
- **BMI above 27–40 kg/m²** → antibody levels ↓ **30–60%** (may be less protective).
- Also give after **platelet transfusion or plasmapheresis** in D-negative women.
- May cause a weakly positive **direct Coombs** in cord blood, but no significant hemolysis.

Large fetomaternal hemorrhage

- **>30 mL whole blood in 2–3 per 1000** pregnancies → one dose is not enough.
- Screening only women with risk factors **misses half** → **screen all D-negative women at delivery**:
 1. **Rosette test** (qualitative): maternal blood + anti-D coats fetal D+ cells; D+ indicator cells added form **rosettes** around fetal cells.
 2. If positive → **quantitative test: Kleihauer-Betke or flow cytometry**.
- **One 300-µg dose per 15 mL fetal red cells (30 mL whole blood)**.
- **IM: max 5 doses in 24 h. IV: 600 µg every 8 h**.
- **Adequate dose** is shown by a **positive indirect Coombs** or fetal cells on rosette testing.

Serological weak D phenotypes (formerly D^u)

- The most common D variants in the US and Europe. Molecular **RHD genotyping** splits them:

Type	Defect	Anti-D immune globulin?
Weak D types 1, 2, 3	Fewer intact D antigens	Not needed — manage as D-positive
Partial D	Protein deletions; D antigen lacks epitopes	Needed (can be sensitized)

- **No molecular testing done** in a serologic weak D woman → **give anti-D**.

5. Fetomaternal Hemorrhage

- Some occurs in all pregnancies; enough to provoke an antigen-antibody reaction in **two thirds**.
- Incidence and volume **rise with gestation** (Fig. 18-2).
- **≥30 mL in ~3 per 1000** pregnancies; **≥150 mL in 1 in 2800** births.
- **Presentation:**
 - **Decreased fetal movement** = most common complaint
 - **Sinusoidal FHR** (infrequent) → evaluate immediately
 - **↑ MCA PSV = most accurate predictor**
 - **Hydrops = ominous**
- ↑ MCA PSV or hydrops → **urgent fetal transfusion or delivery**.
- **No cause in >80%.**
- Quantitative tests don't show **timing or chronicity**.
- **Chronic anemia is better tolerated** (FHR may stay normal until moribund). **Acute** hemorrhage → cerebral hypoperfusion, ischemia, infarction.
- May be found during **stillbirth** evaluation.

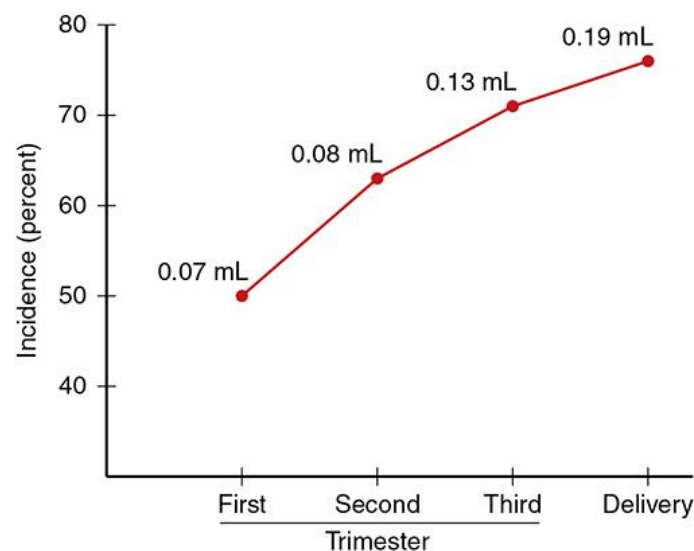


Figure 18-2. Fetomaternal hemorrhage incidence rises from 50% (1st trimester; 0.07 mL) to ~76% at delivery (0.19 mL).

Hemorrhage quantification: Kleihauer-Betke (acid elution)

- **Hb F resists acid elution**; adult Hb A is eluted → fetal cells stain **red**, adult cells appear as **"ghosts"**. Count fetal cells as % of adult cells.

Fetal blood volume = (MBV × maternal Hct × % fetal cells in KB) ÷ newborn Hct

- Term MBV ≈ **5000 mL**.
- **Worked example:** maternal Hct 35%, fetal Hct 50%, KB 1.7% → $5000 \times 0.35 \times 0.017 \div 0.5 = 60$ **mL whole blood**.
 - Fetal-placental volume at term ≈ **125 mL/kg** → 375 mL for 3000 g → fetus lost **~15%**.
 - 60 mL whole blood = **30 mL red cells** → **two 300-μg doses** of anti-D.
- More precise MBV: from maternal height, weight and expected blood volume accrual (Table 42-1).
- **KB inaccurate when:**
 1. **Maternal hemoglobinopathy with ↑ Hb F** (e.g., β-thalassemia)
 2. **At or near term** (fetus already making Hb A)
- **Flow cytometry** (monoclonal anti-Hb F or anti-D, fluorescence): automated, counts more cells, **unaffected** by maternal Hb F or fetal Hb A; **more sensitive and accurate**, but not widely available.

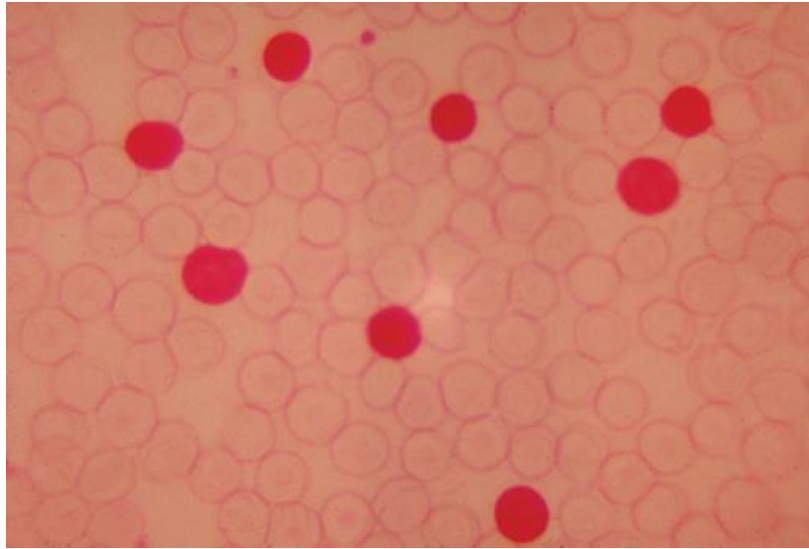


Figure 18-3. Kleihauer-Betke stain: Hb F-rich fetal cells stain darkly; maternal cells are pale "ghosts."

6. Fetal Thrombocytopenia

Fetal and neonatal alloimmune thrombocytopenia (FNAIT / AIT)

- **Most common cause of severe thrombocytopenia in term newborns: 1–2 per 1000 births.**
- Maternal antibodies against **paternally inherited fetal platelet antigens.**
- **Maternal platelet count is normal** (unlike ITP).
- **Can severely affect the FIRST pregnancy** (unlike anti-D).
- **HPA-1a = 80–90%** of cases, and the most severe. Then **HPA-5b > HPA-1b > HPA-3a**. Others 1%.
- **~85%** of non-Hispanic white people are HPA-1a positive; **2% are HPA-1b homozygous** (at risk), but only **10%** of these make antibodies with an HPA-1a fetus.
- **~1/3** of affected fetuses/neonates get severe thrombocytopenia; **10–20%** of those get **ICH**. Population rate: **1 ICH per 25,000–60,000** pregnancies.
- **Spectrum:** incidental thrombocytopenia → petechiae → devastating ICH.
 - Registry of 600: **ICH in 7%; 60% in the first-born; half before 28 wk**; 1/3 died, **50% of survivors severely disabled.**
- **Bussel (1997):** severity predicted by a **prior sibling with ICH**; **50%** had initial platelets **<20,000/μL**; if **>80,000/μL**, counts fell **≥10,000/μL per week** untreated.

Diagnosis and management

- Usually diagnosed after delivery: **unexplained severe neonatal thrombocytopenia + normal maternal platelets**. Rarely after fetal ICH.
- **Recurrers in 70–90%**, often severe and **earlier each pregnancy**.
- Fetal blood sampling (transfuse if **<50,000/μL**) has **>10% complication rate** → **empirical IVIG** is favored.
- **IVIG ↑ platelets ~50,000/μL**; no ICH in 50 treated fetuses (Bussel 1996).
- **+ Corticosteroids** in high risk (platelets **<20,000/μL** or sibling with ICH) → platelet rise in 80%, but a systematic review found **no consistent benefit** over IVIG alone → controversial.
- **Cesarean delivery at or near term** is generally recommended.

Table 18-3. Fetal-Neonatal Alloimmune Thrombocytopenia (FNAIT) Treatment Recommendations

Risk Group	Criteria	Suggested Management
1	Prior fetus or newborn with ICH, but no maternal anti-HPA antibody identified	Maternal anti-HPA antibody screening and cross-matching with paternal platelets at 12, 24 and 32 wk ; no treatment for negative results
2	Prior fetus or newborn with thrombocytopenia and maternal anti-HPA antibody, but no ICH	From 20 wk: IVIG 1 g/kg/wk + prednisone 0.5 mg/kg/d or IVIG 2 g/kg/wk. From 32 wk: IVIG 2 g/kg/wk + prednisone 0.5 mg/kg/d. Continue until delivery
3	Prior fetus with 3rd-trimester ICH or prior newborn with ICH, and maternal anti-HPA antibody	From 12 wk: IVIG 1 g/kg/wk. From 20 wk: either ↑ IVIG to 2 g/kg/wk or add prednisone 0.5 mg/kg/d. From 28 wk: IVIG 2 g/kg/wk and prednisone 0.5 mg/kg/d. Continue until delivery
4	Prior fetus with ICH before the 3rd trimester and maternal anti-HPA antibody	From 12 wk: IVIG 2 g/kg/wk. From 20 wk: add prednisone 1 mg/kg/d . Continue both until delivery

HPA = human platelet antigen; ICH = intracerebral hemorrhage; IVIG = intravenous immunoglobulin G.

Memory hook: The earlier the prior ICH, the earlier (12 wk) and heavier (2 g/kg + prednisone 1 mg/kg) the treatment.

Immune thrombocytopenia (ITP)

- Maternal autoimmune antiplatelet IgG may cross the placenta → **usually mild** fetal thrombocytopenia.
- Neonatal platelets may fall rapidly after birth: **nadir at 48–72 h**.
- **Nothing predicts** fetal/neonatal counts: not maternal platelets, antibodies, or steroid treatment.
- **>400 pregnancies: no fetal/neonatal ICH.**
- **No fetal blood sampling. Mode of delivery by standard obstetric indications.**

Feature	FNAIT	ITP
Maternal platelets	Normal	Low
Antibody against	Paternal platelet antigen (HPA-1a)	Platelet glycoproteins (autoimmune)
Fetal severity	Often severe ; ICH risk, first pregnancy affected	Usually mild ; ICH rare
Management	IVIG ± prednisone ; cesarean near term	No fetal sampling; route by obstetric indications

7. Hydrops Fetalis

- "General dropsy" of the fetus (Ballantyne, 1892).
- **Diagnosis: ≥2 fetal effusions** (pleural, pericardial, ascites) **or 1 effusion + anasarca**.
- **Anasarca** = skin thickness **>5 mm**.
- **Placentomegaly** = thickness **≥4 cm in the 2nd trimester** or **≥6 cm in the 3rd**.
- As hydrops progresses: **anasarca is invariable**, usually with placentomegaly and hydramnios.
- **Immune** (with red cell alloimmunization) vs **nonimmune** (everything else).
- **Shared mechanisms:** ↓ **colloid oncotic pressure**, ↑ **hydrostatic/central venous pressure**, ↑ **vascular permeability**.

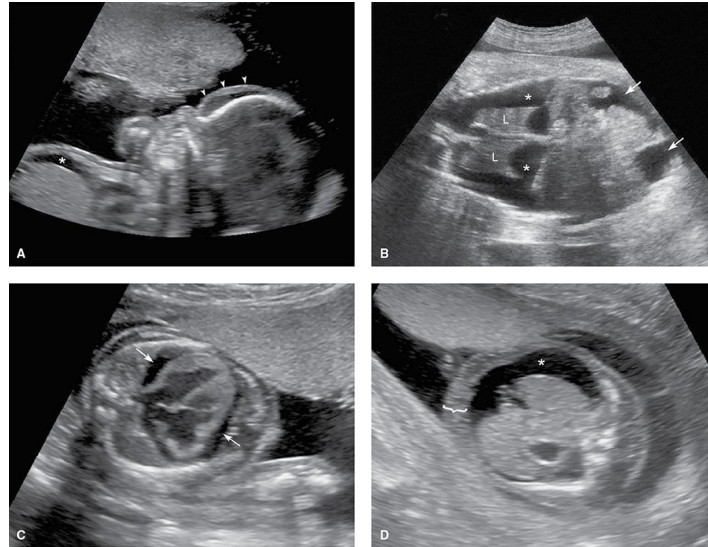


Figure 18-4. Sonographic hydrops: (A) scalp edema + ascites, (B) pleural effusions outlining lungs, (C) pericardial effusion with cardiomegaly, (D) ascites + anasarca with cystic hygroma.

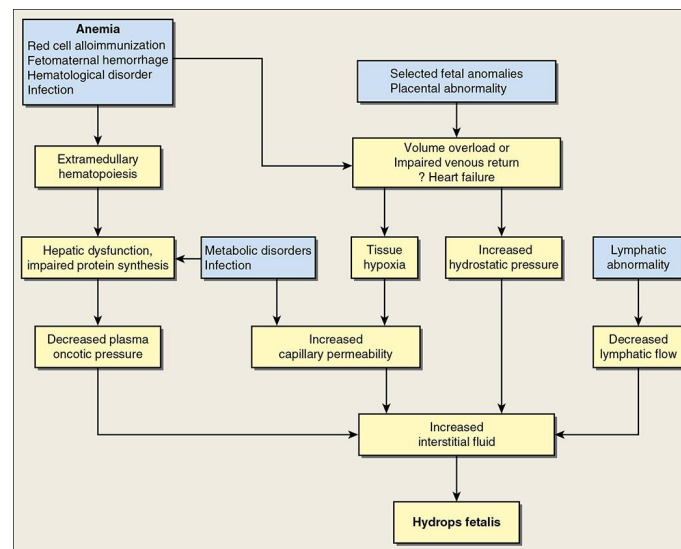


Figure 18-5. Pathogenesis of hydrops: ↓ oncotic pressure, ↑ hydrostatic pressure, ↑ capillary permeability and ↓ lymphatic flow → ↑ interstitial fluid.

Immune hydrops

- Anemia → marrow erythroid hyperplasia + **extramedullary hematopoiesis** (liver, spleen) → **portal hypertension** + ↓ **hepatic protein synthesis** → ↓ **plasma oncotic pressure**.
- Anemia may also ↑ **CVP**; tissue hypoxia ↑ **capillary permeability**.
- Now much rarer (anti-D prophylaxis, MCA Doppler).
- Only severe anemia** causes it: all immune hydrops had **Hb <5 g/dL** (Mari 2000). **Immediate fetal transfusion** may be lifesaving.

Nonimmune hydrops

- ≥90% of hydrops. ~1 per 1500 second-trimester pregnancies.
- Cause found: ~60% prenatally**, up to **80% postnatally**.
- Prenatal causes: **aneuploidy ~20%**, **cardiovascular 15%**, **infection 14%** (most commonly **parvovirus B19**).
- Twins: TTTS** is the most frequent cause.
- Prognosis:** only **40%** result in a liveborn, and of these **neonatal survival ~50%**.
- Cause varies with GA at diagnosis:**
 - 1st trimester: aneuploidy 70%**

- **14–24 wk:** aneuploidy and infection **20% each**
- **<24 wk:** commonest aneuploidy is **45,X (Turner)**; survival **<5%**
- **Whole exome sequencing** in unexplained cases (127): **~30%** pathogenic variant; commonest = **RASopathies (Noonan)**.
- **Thailand/Southern China:** **α^0 -thalassemia = 30–50%**, extremely poor prognosis.
- **Treatable causes** (parvovirus, chylothorax, tachyarrhythmias; **~10% each**) → **survival in ⅓** with fetal therapy.

Table 18-4. Categories and Etiologies of Nonimmune Hydrops Fetalis

Category	Examples	Percent ^a
Cardiovascular	Structural defects: Ebstein anomaly, tetralogy of Fallot with absent pulmonary valve, hypoplastic left or right heart, premature closure of ductus arteriosus, arteriovenous malformation (vein of Galen aneurysm); cardiomyopathies; tachyarrhythmias; bradycardia, as in heterotaxy syndrome with endocardial cushion defect or with anti-Ro/La antibodies	21
Chromosomal	Turner syndrome (45,X), triploidy, trisomies 21, 18 and 13	13
Hematological	Hemoglobinopathies, such as α^0 -thalassemia; erythrocyte enzyme and membrane disorders; erythrocyte aplasia/dyserythropoiesis; decreased erythrocyte production (myeloproliferative disorders); fetomaternal hemorrhage	10
Lymphatic abnormalities	Cystic hygroma, systemic lymphangiectasis, pulmonary lymphangiectasis	8
Infections	Parvovirus B19, syphilis, CMV, toxoplasmosis, rubella, enterovirus, varicella, herpes simplex, coxsackievirus, listeriosis, leptospirosis, Chagas disease, Lyme disease	7
Syndromic	Arthrogryposis multiplex congenita, lethal multiple pterygium, congenital lymphedema, myotonic dystrophy type I, Neu-Laxova, Noonan and Pena-Shokeir syndromes	5
Thoracic abnormalities	Cystic adenomatoid malformation, pulmonary sequestration, diaphragmatic hernia, hydro/chylothorax, CHAOS, mediastinal tumors, skeletal dysplasia with very small thorax	5
Gastrointestinal	Meconium peritonitis, GI tract obstruction	1
Kidney and urinary tract	Kidney malformations, bladder outlet obstructions, congenital (Finnish) nephrosis, Bartter syndrome, mesoblastic nephroma	2
Placental, twin and cord	Placental chorioangioma, TTTS, TRAP sequence, twin anemia polycythemia sequence, cord vessel thrombosis	5
Other rare disorders	Inborn errors of metabolism (Gaucher, galactosialidosis, GM ₁ gangliosidosis, sialidosis, mucopolysaccharidoses, mucopolipidoses); tumors (sacrocccygeal teratoma, hemangioendothelioma with Kasabach-Merritt syndrome)	5
Idiopathic	—	18

^aPercentages reflect the proportion within each category from a systematic review of 6775 pregnancies with nonimmune hydrops. CHAOS = congenital high airway obstruction sequence; TRAP = twin reversed arterial perfusion. The printed table reads " α^0 -thalassemia", an evident typo for α^0 -thalassemia (the term used in the chapter text).

Memory hook: Top three identifiable categories: **"Heart, Chromosomes, Blood"** (21%, 13%, 10%), with **idiopathic 18%**.

Diagnostic evaluation

1. **Indirect Coombs** (alloimmunization)
2. **Detailed ultrasound** of fetus and placenta:
 - Anatomical survey (Table 18-4 structural causes)
 - **Fetal echocardiography**
 - **MCA PSV** for anemia

3. **Amniocentesis: chromosomal microarray or karyotype**, plus **parvovirus B19, CMV, toxoplasmosis** testing
4. **Kleihauer-Betke** if anemia suspected
5. Consider **α -thalassemia** and/or **inborn errors of metabolism** testing
 - **Whole exome sequencing** if the above is uninformative. Counsel about turnaround, cost and **variants of uncertain significance; not for routine use** (ACOG). Trio sequencing can reveal incidental parental findings (e.g., cancer predisposition).

Isolated effusion or edema

- Not diagnostic of hydrops, but do the evaluation and surveil closely.

Finding	Think of
Pericardial effusion	May precede hydrops in parvovirus B19
Isolated ascites	Parvovirus B19; meconium peritonitis
Isolated pleural effusion	Chylothorax — fetal therapy can be lifesaving if hydrops develops
Edema of upper torso, dorsum of hands/feet	Turner or Noonan syndrome; congenital lymphedema

8. Mirror Syndrome (Ballantyne, "triple edema")

- **Mother mirrors the hydropic fetus**: fetus, mother and placenta all edematous.
- Complicates **$\geq 20\%$** of hydrops cases. **Etiology of the hydrops does not matter** (D alloimmunization, TTTS, chorioangioma, cystic hygroma, Ebstein, SCT, chylothorax, bladder outlet obstruction, SVT, vein of Galen, infections).
- **Maternal findings (Braun, >50 cases)**:

Finding	Frequency
Edema	~90%
Hypertension	60%
Proteinuria	40%
$\uparrow$ Liver enzymes	20%
Headache, visual disturbance	~15%

- Considered a **form of severe preeclampsia**: same angiogenic imbalance ($\uparrow$ **sFlt-1**, $\downarrow$ **PIGF**, $\uparrow$ **sVEGFR1**). Others argue it is distinct, with **hemodilution** rather than hemoconcentration.
- **Management: usually prompt delivery** $\rightarrow$ maternal findings resolve.
- Successful **fetal therapy** (anemia, SVT, hydrothorax, bladder outlet obstruction, parvovirus transfusion) has reversed both hydrops and mirror syndrome, but delay delivery only **with caution; deliver if the mother deteriorates**.

Quick-Recall: Antibodies, Tests and Thresholds

Item	Key fact
Anti-D, anti-Kell, anti-c, anti-E	Cause most severe fetal anemia needing transfusion
Anti-Kell	Marrow suppression; titer unreliable → MCA Doppler regardless; 90% from transfusion
Anti-E vs anti-c	E most common; c more often needs transfusion
Lewis, I, Fy ^b	No fetal hemolysis
ABO	Most common newborn hemolysis; firstborn; mild; no fetal surveillance
Indirect Coombs	Free antibody in maternal serum (screen; also weakly + 1:1–1:4 after anti-D)
Critical titer	1:8–1:32 (often 1:16); repeat every 4 wk below it; don't repeat once reached
MCA PSV	>1.5 MoM = moderate–severe anemia (1.70 MoM after first transfusion); unreliable >34–35 wk
Rosette test	Qualitative screen for fetal D+ cells at delivery
Kleihauer-Betke	Quantifies FMH; inaccurate with maternal ↑ Hb F or near term
Flow cytometry	More accurate than KB; not widely available
Amniocyte PCR	Fetal genotype: PPV 100%, NPV ~97%; avoid CVS
cfDNA RHD	Sens >99%, spec >95%; not routine in US
FNAIT	HPA-1a (80–90%); normal maternal platelets; 1st pregnancy; IVIG ± prednisone
ITP	Mild fetal effect; neonatal nadir 48–72 h; no fetal sampling

Key Numbers to Memorize

- Alloimmunization **~1%** of pregnancies; D sensitization from **0.1 mL**; RH genes on **chromosome 1p**
- No prophylaxis → **16%** sensitized (2% / 7% / 7%); ABO-incompatible → **2%**
- Anti-D **300 µg = 30 mL whole blood / 15 mL red cells**; at **28 wk + ≤72 h postpartum**; half-life **16–24 d**; protection up to **28 d**; 1st tri **50–120 µg**
- Prophylaxis: postpartum ↓ risk **90%**; 28-wk dose **2% → 0.1%**; overall **<0.2%**
- FMH **>30 mL in 2–3/1000**; **≥150 mL in 1/2800**; no cause in **>80%**
- KB: **MBV × maternal Hct × % fetal cells ÷ newborn Hct**; MBV **5000 mL**; fetoplacental volume **125 mL/kg**
- Transfuse if **Hct <30%**; target **40–50%**; donor Hct **~80%**; volume **EFW × 0.02 per 10%**; Hct falls **1%/day**; repeat every **2–4 wk**
- IUT survival **~95%**; hydrops **~80%**; fetal death **2%**; <20 wk stillbirth **>15%**
- FNAIT **1–2/1000**; HPA-1a **80–90%**; recurrence **70–90%**; IVIG ↑ platelets **~50,000/µL**
- Hydrops: **≥2 effusions** or 1 + anasarca; skin **>5 mm**; placenta **≥4 cm (2nd) / ≥6 cm (3rd)**
- Immune hydrops: **Hb <5 g/dL**; nonimmune **≥90%**, **1/1500**; liveborn **40%**, neonatal survival **50%**; Turner <24 wk survival **<5%**
- Mirror syndrome **≥20%** of hydrops; maternal edema **90%**, HTN **60%**